

Assignment 1

Problems:

1. Which of the following expressions are statements?
 - (a) Today is a nice day.
 - (b) Go to sleep.
 - (c) Is it going to snow tomorrow?
 - (d) The U.S. has 49 states.
 - (e) I like to eat fruit, and you often think about traveling to Spain.
 - (f) If we go out tonight, the babysitter will be unhappy.
 - (g) Call me on Thursday if you are home.
2. Let X = “I am happy,” let Y = “I am watching a movie” and Z = “I am eating spaghetti.” Translate the following statements into words.
 - (a) $Z \rightarrow X$.
 - (b) $X \leftrightarrow Y$.
 - (c) $(Y \vee Z) \rightarrow X$.
 - (d) $Y \vee (Z \rightarrow X)$.
 - (e) $(Y \rightarrow \neg X) \wedge (Z \rightarrow \neg X)$.
 - (f) $(X \wedge \neg Y) \leftrightarrow (Y \vee Z)$.
3. Let E = “The house is blue,” let F = “The house is 30 years old” and G = “The house is ugly.” Translate the following statements into symbols.
 - (a) If the house is 30 years old, then it is ugly.
 - (b) If the house is blue, then it is ugly or it is 30 years old.
 - (c) If the house is blue, then it is ugly, or it is 30 years old.
 - (d) The house is not ugly if and only if it is 30 years old.
 - (e) The house is 30 years old if it is blue, and it is not ugly if it is 30 years old.
 - (f) For the house to be ugly, it is necessary and sufficient that it be ugly and 30 years old.
4. Make a truth table for each of the following statements.
 - (a) $P \wedge \neg Q$.
 - (b) $(R \vee S) \wedge \neg R$.
 - (c) $X \vee (\neg Y \vee Z)$.
 - (d) $(A \vee B) \wedge (A \vee C)$.
 - (e) $(P \wedge R) \vee \neg(Q \wedge S)$.

5. Simplify the following statements (making use of any equivalences of statements given so far in the text or exercises).

- (a) $\neg(P \rightarrow \neg Q)$
- (b) $A \rightarrow (A \wedge B)$
- (c) $(X \wedge Y) \rightarrow X$
- (d) $\neg(M \vee L) \wedge L$
- (e) $(P \rightarrow Q) \vee Q$
- (f) $\neg(X \rightarrow Y) \vee Y$

6. Suppose that the possible values of x are all people. Let $Y(x) = "x \text{ has green hair}"$, let $Z(x) = "x \text{ likes pickles}"$ and $W(x) = "x \text{ has a pet frog}"$. Translate the following statements into words.

- (a) $(\forall x)Y(x)$.
- (b) $(\exists x)Z(x)$.
- (c) $(\forall x)[W(x) \wedge Z(x)]$.
- (d) $(\exists x)[Y(x) \rightarrow W(x)]$.
- (e) $(\forall x)[W(x) \leftrightarrow \neg Z(x)]$.